COMPSCI1: Homework 2

Programming Concepts and Methodology I

Project 2 - Game of Nim

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Pseudocode

1. Start Game
    a. Set seed of random number generator to current time
2. Print Game Introduction
    a. Display an introduction of the game
    b. Display the rules of the game
        i. Pile starts with 10-21 stones (randomly generated)
        ii. Each turn, player alternates removing 1-3 stones
        iii. Last player to make a move loses
3. Print statement prompting human player to choose between easy ("e") or hard mode ("h")
4. Read human player input
    a. If human player input is invalid (not "e" or "h") print warning/error message and ask for human player input again
    b. Read human player input again (repeat until input is valid)
5. If statement: if human player input "e" then easy mode is implemented (set mode = easy)
6. Else if statement: if human player input "h" or then hard mode is implemented (set mode = hard)
7. Randomly generate start pile size between 10-21 stones
    a. Print number of starting stone pile
8. Randomly choose what player goes first, human player or computer (50-50 chance)
    a. Generate random number 0 or 1
    b. If statement:
        i. If number is 0, human player starts (set player turn = 0 for human)
    c. Else:
        i. If number is 1, computer starts (set player turn = 1 for computer)
    d. Print which player will start (human or computer)
9. Define functions
    a. Easy mode function:
        i. Chooses random number of stones 1-3, but not more than stones left, for computer turn to remove
    b. Hard mode function:
        i. k * 4 + 1 to decide number of stones for computer turn to remove if not already k * 4 + 1 strategy
10. While statement:
    a. While pile of stones is greater than 0
        i. If statement:
            1. If player turn = 0 (human player's turn), ask how many stones to remove (1-3)
            2. Read human player input
            3. While statement:

            a. While input is less than 1, greater than 3, or greater than stones left (should not have negative stones), print a warning that input is invalid and ask for input again
            b. Ask how many stones to remove again (1-3)
            c. Read human player input again
        4. Remove the number of stones inputted
        5. State number of stones human player removed
        6. State number of stones left
        7. If statement:
            a. If stones = 0, state that the human player removed the last stone and lost
            b. Exit while loop
        8. Switch turn to the computer (set player turn = 1)
    ii. Else if statement:
        1. If player turn = 1 (computer's turn),
            a. If statement:
                i. If mode is easy, computer randomly generates/chooses a number between 1-3
                ii. If statement:
                1. If number chosen is greater than stones left, lower the number to equal the number of stones left
            b. Else statement:
                i. Else if mode is hard:
                1. If statement:
                a. If pile size is 1, 5, 9, 13, 17, or 21 (multiple of four plus 1), computer removes 1 stone
                2. Else statement:
                a. Else remove number of stones that leaves the pile at 1, 5, 9, 13, 17, or 21 (k * 4 + 1)
        2. Remove the number of stones computer chose
        3. State number of stones computer removed
        4. State the number of stones left
        5. If statement:
            a. If stones = 0, state that the computer removed the last stone and human player won
            b. Exit while loop
        6. Switch turn to the human player (set player turn = 0)

11. Display “********** Thank you for playing **********”
12. End Game

Screenshots

Easy Game Mode

- Includes example of invalid input for game mode
- Includes example of me losing output

```cpp
/*
Filename: luis_ceja_abrica_project2.cpp
Description: C++ program foor a variant of Nim with easy mode where the computer player randomly chooses 1-3 stones to remove from the pile while never allowing the number of stones left to be negative and hard mode where the computer player will always try to win when feasible.
Author: Luis Eduardo Ceja Abrica
Class: Programming Concepts and Methodology I
Date: November 22, 2025
*/

#include <iostream>
#include <cstdlib> /* for rand and srand*/
#include <ctime> /* to make it truly random at time 0 for current time */
#include <unistd.h> /* for sleep extra credit, did not do Windows.h as I have a Mac*/

using namespace std;


/* setting constants for our min and max number of stones we are
```

```
********** THE GAME OF NIM **********

Rules of the game:
* Each player can remove 1, 2, or 3 stones from the pile.
* The player who makes the last move loses.

Choose mode:
Easy Mode (e) or Hard Mode (h): 8
Warning, please enter 'e' or 'h': k
Warning, please enter 'e' or 'h': E
Warning, please enter 'e' or 'h': e
You have chosen the easy mode.

The initial number of stones is 13
Now a roll of the die to determine who plays first...
You play first.

It is your turn. Enter the number of stones you would like to remove: 3
You removed 3 stones from the pile
The pile now has 10 stones left.

The computer is choosing a move...
The computer removed 2 stones from the pile
```

```cpp
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```

```
The computer removed 2 stones from the pile
The pile now has 8 stones left.

It is your turn. Enter the number of stones you would like to remove:
3
You removed 3 stones from the pile
The pile now has 5 stones left.

The computer is choosing a move...
The computer removed 3 stones from the pile
The pile now has 2 stones left.

It is your turn. Enter the number of stones you would like to remove: 2
You removed 2 stones from the pile
The pile now has 0 stones left.

The pile now has 0 stones left.

Sorry, you lost the game :(

********** Thank you for playing **********
```

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/* setting constants for our min and max number of stones we are
```

```
It is your turn. Enter the number of stones you would like to remove:
3
You removed 3 stones from the pile
The pile now has 5 stones left.

The computer is choosing a move...
The computer removed 3 stones from the pile
The pile now has 2 stones left.

It is your turn. Enter the number of stones you would like to remove: 2
You removed 2 stones from the pile
The pile now has 0 stones left.

The pile now has 0 stones left.

Sorry, you lost the game :(

************** Thank you for playing **************


=== Code Execution Successful ===
```

Hard Game Mode

- Includes example of invalid move
- Includes example of me winning output

```cpp
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using namespace std;


/* setting constants for our min and max number of stones we are
```

```
************** THE GAME OF NIM **************

Rules of the game:
* Each player can remove 1, 2, or 3 stones from the pile.
* The player who makes the last move loses.

Choose mode:
Easy Mode (e) or Hard Mode (h): h
You have chosen the hard mode.

The initial number of stones is 14
Now a roll of the die to determine who plays first...
You play first.

It is your turn. Enter the number of stones you would like to remove: 6
6 is not a valid move. Enter either 1, 2, or 3 for the number of stones: 5
5 is not a valid move. Enter either 1, 2, or 3 for the number of stones: 0
0 is not a valid move. Enter either 1, 2, or 3 for the number of stones: 3
You removed 3 stones from the pile
The pile now has 11 stones left.

The computer is choosing a move...
The computer removed 3 stones from the pile
```

```cpp
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using namespace std;


/* setting constants for our min and max number of stones we are
```

```
The computer is choosing a move...
The computer removed 3 stones from the pile
The pile now has 8 stones left.

It is your turn. Enter the number of stones you would like to remove: 3
You removed 3 stones from the pile
The pile now has 5 stones left.

The computer is choosing a move...
The computer removed 1 stone from the pile
The pile now has 4 stones left.

It is your turn. Enter the number of stones you would like to remove: 1
You removed 1 stone from the pile
The pile now has 3 stones left.

The computer is choosing a move...
The computer removed 3 stones from the pile
The pile now has 0 stones left.

The pile now has 0 stones left.
```

Visual Studio


```cpp
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using namespace std;


/* setting constants for our min and max number of stones we are
```


```
The pile now has 5 stones left.

The computer is choosing a move...
The computer removed 1 stone from the pile
The pile now has 4 stones left.

It is your turn. Enter the number of stones you would like to remove: 1
You removed 1 stone from the pile
The pile now has 3 stones left.

The computer is choosing a move...
The computer removed 3 stones from the pile
The pile now has 0 stones left.

The pile now has 0 stones left.

Congratulations, you won! :)

************** Thank you for playing **************


=== Code Execution Successful ===
```

Visual Studio